
Hyperbolic JEPA: Poincaré Predictors Compose with Sketched Anti-Collapse for Self-Supervised Pretraining

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Abstract

Self-supervised joint-embedding predictive architectures (JEPAs) learn representations by predicting masked feature embeddings in a flat Euclidean space. We ask a narrow but load-bearing question: does the *geometry of the prediction loss* matter, given modern anti-collapse regularizers? We introduce a drop-in replacement for I-JEPA’s Euclidean prediction loss: a Poincaré-ball predictor with curvature $c=1$ applied on top of an unchanged Euclidean encoder. We study how it composes with VICReg and with SIGReg (LeJEPA’s Sketched Isotropic Gaussian Regularizer). Across CIFAR-10, CIFAR-100, and Tiny-ImageNet with a ViT-Tiny backbone and three seeds per cell, the Poincaré predictor outperforms the Euclidean baseline in **all twelve** dataset- $\times$ -regularizer cell means. At Tiny-ImageNet scale the benefit compounds with a tangent-space formulation of SIGReg: the gap between B+SIGReg and B+VICReg widens from +2.9 pp at 300 epochs to +4.8 pp at 600 epochs, with clean per-seed separation at 600 epochs. We also report two secondary findings: vanilla Euclidean JEPA exhibits *negative* scaling on Tiny-ImageNet (probe accuracy drops by 2.3 pp on every seed when training doubles from 300 to 600 epochs), and an apparent catastrophic collapse of ambient SIGReg on small ViTs is a regularizer-weight issue (not an architectural limitation) rescued by a $10\times$ weight reduction.

1 Introduction

Joint-embedding predictive architectures (JEPAs) [LeCun, 2022, Assran et al., 2023] learn representations by predicting masked feature embeddings of an image’s target patches from its visible context patches. Unlike pixel-space masked autoencoders [He et al., 2022] or contrastive methods that compare pairs of augmentations [Chen et al., 2020, Caron et al., 2021], JEPAs operate entirely in feature space: a predictor network maps context embeddings to target embeddings under an ℓ_2 loss. The design trades pixel-level reconstruction fidelity for semantic prediction.

A recurring concern with feature-space SSL is representation *collapse*: the trivial solution of a constant representation makes the prediction loss zero. Prior work addresses collapse through asymmetric architectures [Grill et al., 2020, Chen and He, 2021], clustering bottlenecks [Caron et al., 2020], or explicit anti-collapse regularizers such as Barlow Twins [Zbontar et al., 2021], VICReg [Bardes et al., 2022], and, most recently, LeJEPA’s Sketched Isotropic Gaussian Regularizer (SIGReg) [Balestriero and LeCun, 2025]. These regularizers all operate on Euclidean feature vectors and shape the marginal distribution of the learned representation.

In parallel, a line of work has argued that visual data has latent hierarchical or tree-like structure that is embedded more efficiently in hyperbolic space than in Euclidean space [Nickel and Kiela, 2017, Khrulkov et al., 2020, Ganea et al., 2018]. Follow-up work has built hyperbolic counterparts of image

classifiers [van Spengler et al., 2023], contrastive SSL [Yue et al., 2023], and transformers [Chen et al., 2022]. These approaches range from full-network hyperbolic models, which add optimization difficulty and break compatibility with pretrained Euclidean encoders, to lighter-weight designs that keep the encoder Euclidean and curve only an output head or loss; our work belongs to the second category, applied to JEPA-style masked-feature prediction.

We ask whether the benefit of curvature can be captured with a much smaller intervention: curving only the prediction loss while leaving the encoder Euclidean. Concretely, we replace I-JEPA’s ℓ_2 loss with a Poincaré-ball distance [Ganea et al., 2018] computed between predictor outputs and target embeddings projected onto the ball. This is a one-line change in the loss function. We then study how this geometric change interacts with the two dominant anti-collapse regularizers used in recent SSL work (VICReg and SIGReg), and how the interaction scales with compute.

Our empirical study is small but deliberately thorough. We run a $2 \times 2 \times 3$ matrix (Euclidean vs. Poincaré predictor, $\times$ VICReg vs. SIGReg, $\times$ three datasets) with three seeds per cell, identical backbones and training budgets, and a set of focused follow-up experiments to address scale, projector effects, hyperparameters, and curvature.

Contributions.

1. **Poincaré predictor (Model B).** A drop-in replacement for I-JEPA’s Euclidean prediction loss that curves only the output head, not the encoder. This modification wins in *all twelve* dataset- $\times$ -regularizer cells of our main experimental matrix (Section 5.1); margins range from a tight +0.3 pp (Tiny-ImageNet, SIGReg with the λ -rescued baseline) to +10.8 pp (CIFAR-100, SIGReg), and the gap widens with compute (Section 5.2).
2. **Tangent-space SIGReg.** Applying SIGReg to Poincaré-ball features requires lifting them to the tangent space at the origin via the logarithmic map; we show this is the formulation that composes with the hyperbolic predictor (Section 3.2).
3. **Scale-dependent regularizer choice.** At small scale, VICReg wins; at Tiny-ImageNet with 600 epochs, SIGReg wins with clean seed separation and the gap widens with compute (Section 5.2).
4. **Negative scaling of Euclidean JEPA.** Doubling training from 300 to 600 epochs causes the probe accuracy of the Euclidean baseline (A+VICReg) to *decrease* by 2.3 pp (mean across seeds, with all three seeds regressing), while B+SIGReg *gains* 2.8 pp over the same compute budget (Sections 5.2, 5.3).
5. **A hyperparameter fix for ambient SIGReg on small ViTs.** Naively applied, ambient SIGReg fails to learn useful representations at Tiny-ImageNet scale (1.8% linear probe). We show this is a regularizer-*weight* issue rather than a learning-rate issue: reducing λ_{sig} from 0.1 to 0.01 recovers 13.7%, stable across seeds (Section 5.4).
6. **Projector- $\times$ -regularizer interaction.** The batch-normalization projector recommended by LeJEPA *helps* SIGReg by +2.7 pp but *hurts* VICReg by -3.6 pp, and this flips the regularizer ordering at 300 epochs (Section 5.5). We disclose this interaction explicitly rather than selecting the most favorable variant.

2 Background

2.1 I-JEPA

I-JEPA [Assran et al., 2023] learns image representations by predicting the feature-space embeddings of masked “target” patches from those of visible “context” patches. Let f_θ be an encoder ViT and g_ϕ a predictor network. For an image x with context mask M_{ctx} and a family of target masks $\{M_{\text{tgt}}^{(k)}\}_k$, I-JEPA uses the loss

$$\mathcal{L}_{\text{pred}}(\theta, \phi) = \mathbb{E}_{x,k} \|g_\phi(f_\theta(x \odot M_{\text{ctx}})) - \text{sg}(f_\theta(x \odot M_{\text{tgt}}^{(k)}))\|_2^2, \quad (1)$$

where sg is the stop-gradient and $\bar{\theta}$ is an exponential-moving-average (EMA) copy of θ . The squared- ℓ_2 prediction loss in Eq. 1 is the geometric object we target in this paper.

2.2 Anti-collapse regularizers

VICReg [Bardes et al., 2022] enforces two structural properties on a batch of embeddings $Z \in \mathbb{R}^{B \times d}$: a per-dimension variance hinge $V(Z) = \frac{1}{d} \sum_{j=1}^d \max(0, \gamma - \sqrt{\text{Var}(Z_{:,j}) + \epsilon})$, and an off-diagonal covariance penalty $C(Z) = \frac{1}{d} \sum_{i \neq j} [\text{Cov}(Z)]_{ij}^2$. Together with an invariance term across two views, the VICReg loss shapes the first two moments of the representation. In the JEPA setting we have a single (encoder) view per image rather than two augmented views, so we drop the invariance term and combine V and C with the canonical VICReg ratio 25:1 [Bardes et al., 2022]; the predictor–target matching that I-JEPA already performs plays the role the invariance term would otherwise play.

SIGReg [Balestriero and LeCun, 2025] takes a different approach. The LeJEPA authors argue that the *optimal* anti-collapse target is an isotropic Gaussian: it simultaneously maximizes entropy, achieves zero off-diagonal covariance, and has full effective rank. SIGReg tests whether the distribution of Z matches $\mathcal{N}(0, I)$ via an Epps–Pulley normality test [Epps and Pulley, 1983] applied to M random one-dimensional slices:

$$\mathcal{L}_{\text{sig}}(Z) = \frac{1}{M} \sum_{m=1}^M \int_0^{t_{\max}} |\hat{\phi}_m(t) - e^{-t^2/2}|^2 e^{-t^2/2} dt, \quad (2)$$

where $\hat{\phi}_m(t)$ is the empirical characteristic function of $Z u_m$ for u_m uniformly distributed on S^{d-1} . The integral is evaluated on a fixed quadrature with n_{points} nodes; the integrand is symmetric in t so we integrate over $[0, t_{\max}]$ rather than the symmetric interval used in the LeJEPA reference code. In our notation, we use $M=256$, $t_{\max}=3$, and $n_{\text{points}}=17$; n_{points} matches the LeJEPA reference, while M and $t_{\max}$ are within the operating ranges explored in the LeJEPA Table 1a ablation.

2.3 Poincaré ball

The Poincaré ball of curvature $c > 0$ is the open ball $\mathbb{D}_c^d = \{z \in \mathbb{R}^d : c\|z\|^2 < 1\}$ equipped with the Riemannian metric $g_z^c = \lambda_z^2 g^E$ where $\lambda_z = 2/(1 - c\|z\|^2)$ and g^E is the Euclidean metric. We will use the geodesic distance, the logarithmic map at the origin, and the radius-clipping operation:

$$d_c(u, v) = \frac{2}{\sqrt{c}} \text{artanh}(\sqrt{c} \|(-u) \oplus_c v\|), \quad (3)$$

$$\log_0^c(z) = \frac{1}{\sqrt{c}} \text{artanh}(\sqrt{c} \|z\|) \cdot \frac{z}{\|z\|}, \quad (4)$$

$$\text{clip}_r(z) = \begin{cases} z & \text{if } \|z\| \leq r, \\ r \cdot z / \|z\| & \text{otherwise.} \end{cases} \quad (5)$$

Here $\oplus_c$ is the Möbius addition [Ganea et al., 2018]. To avoid the boundary singularity where $\lambda_z \rightarrow \infty$, we apply two safeguards: (i) the Euclidean predictor and target are clipped to radius $r_{\max} = 1.0$ before being lifted to the ball (Sec. 3.1), and (ii) any internal point with $\sqrt{c}\|z\| \geq 1 - 10^{-5}$ is projected back to $(1 - 10^{-5})/\sqrt{c}$ inside the ball operations (log map, Möbius addition, distance).

3 Method

3.1 Euclidean vs. Poincaré prediction loss

Denote the predictor output $\hat{z} = g_\phi(f_\theta(x \odot M_{\text{ctx}}))$ and the target $z = \text{sg}(f_{\hat{\theta}}(x \odot M_{\text{tgt}}))$. Both live in $\mathbb{R}^d$. We consider two prediction losses.

Model A (Euclidean, baseline). The standard I-JEPA loss with an elementwise reduction $\ell^A(\hat{z}, z) = \frac{1}{pd} \sum_{j=1}^d |\hat{z}_j - z_j|^p$, with $p=2$. For $p=2$ this equals $(1/(2d)) \|\hat{z} - z\|_2^2$, i.e. a half-MSE per coordinate, and is Eq. 1 up to constants.

Model B (Poincaré). We clip $\hat{z}$ and z to the Euclidean ball of radius $r_{\max}=1$, lift them to the Poincaré ball via the exponential map at the origin, and evaluate the geodesic distance:

$$\ell^B(\hat{z}, z) = d_c(\exp_0^c(\text{clip}_{r_{\max}}(\hat{z})), \exp_0^c(\text{clip}_{r_{\max}}(z)))^p. \quad (6)$$

The encoder and predictor remain standard (Euclidean-output) transformer blocks; only the loss is curved. In particular, model B is a drop-in replacement that does not require hyperbolic-aware layers [Ganea et al., 2018, van Spengler et al., 2023], does not change the number of parameters, and does not change downstream probe compatibility.

3.2 Tangent-space SIGReg

SIGReg’s target distribution is $\mathcal{N}(0, I)$ on $\mathbb{R}^d$. If the regularized features live on the Poincaré ball, a point near the boundary contributes a vanishing metric ball (the conformal factor λ_z blows up), so a Gaussian target in the raw ball coordinates is mis-specified: the regularizer would push features toward a distribution whose density on the manifold is concentrated at the origin. We resolve this by regularizing in the *tangent space at the origin*: given a batch Z on the ball, compute $\tilde{Z}_i = \log_0^c(Z_i)$ (Eq. 4), then apply the standard ambient SIGReg to $\tilde{Z}$.

$$\mathcal{L}_{\text{tan-sig}}(Z; c) = \mathcal{L}_{\text{sig}}(\{\log_0^c(Z_i)\}_{i=1}^B). \quad (7)$$

Tangent-space SIGReg is the only variant we use in combination with Model B. For Model A we always use ambient SIGReg applied directly to the Euclidean features, as in the original LeJEPa formulation.

3.3 Batch-normalization projector

The Vision Transformer architecture applies LayerNorm at its output, which centres and rescales each sample’s feature vector and eliminates the cross-batch distributional signal that SIGReg (and to a lesser extent VICReg) operates on. We address this by inserting a small BatchNorm projector before the regularizer in all “+proj” configurations: a single linear layer followed by BatchNorm, $z \mapsto \text{BN}(Wz)$, with output dimension equal to the input dimension ($d=192$). We attach it after the encoder output for the regularizer branch only; the prediction loss is always computed on the raw (pre-projector) embeddings. We treat the choice between BN-projected and unprojected features as a hyperparameter and ablate it in Sec. 5.5.

4 Experimental setup

Datasets. CIFAR-10 (50 epochs) and CIFAR-100 (200 epochs) at their native 32×32 resolution; Tiny-ImageNet [Le and Yang, 2015] (300 or 600 epochs) at 64×64 resolution with 8×8 patches. All datasets use standard train/val splits; no extra augmentation beyond random crops and flips.

Architecture. ViT-Tiny [Dosovitskiy et al., 2021] encoder ($d=192$, 12 layers, 3 heads); predictor is a smaller ViT with $d_{\text{pred}}=96$, 4 layers, 4 heads, following I-JEPA’s default configuration. When the BN projector is used, its input and output dimensions match the encoder output ($d=192$).

Masking. I-JEPA-style: context blocks sample 85–100% of the image, target blocks sample 15–20% with aspect ratio in $[0.75, 1.5]$. Four target blocks per image per step, no overlap with the context block. We use `min_keep=4` patches.

Optimization. AdamW with cosine learning-rate schedule; start $\eta=2 \times 10^{-4}$, final $\eta=10^{-6}$. Peak learning rate, warmup length, and final weight-decay vary by training budget (CIFAR-10/50ep: peak $\eta=8 \times 10^{-4}$, 3-epoch warmup, weight decay $0.04 \rightarrow 0.2$; CIFAR-100/200ep: peak $\eta=10^{-3}$, 10-epoch warmup, weight decay $0.04 \rightarrow 0.4$; Tiny-ImageNet/300ep: peak $\eta=10^{-3}$, 15-epoch warmup, weight decay $0.04 \rightarrow 0.4$; Tiny-ImageNet/600ep: peak $\eta=10^{-3}$, 30-epoch warmup, weight decay $0.04 \rightarrow 0.4$). EMA target encoder with $\tau=0.996 \rightarrow 1.0$. Mixed-precision training (fp16 with gradient scaling). Gradient clip 3.0. Batch size 256. Regularizer weight $\lambda_{\text{reg}}=0.1$ unless otherwise stated.

Regularizer-specific settings. VICReg: variance hinge $\gamma=1$, variance weight 25 and covariance weight 1 (the canonical VICReg ratio). SIGReg: $M=256$ slices, $t_{\text{max}}=3$, $n_{\text{points}}=17$, weight $\lambda_{\text{sig}}=0.1$ (baseline) or 0.01 (low-weight variant, Sec. 5.4). The overall regularizer term is multiplied by $\lambda_{\text{reg}}=0.1$ before being added to the prediction loss.

Table 1: **Main matrix.** Linear-probe accuracy (%) on each dataset’s val set. All runs use the BN projector, $\lambda_{\text{reg}}=0.1$, $c=1.0$ for Model B, and three seeds. **Bold** marks the better of (A, B) within each regularizer $\times$ dataset cell.

Dataset	VICReg			SIGReg		
	A	B	Δ	A	B	Δ
CIFAR-10 (50 ep)	35.2	43.7	+8.5	29.5	34.4	+4.9
CIFAR-100 (200 ep)	18.0	19.6	+1.6	8.6 [†]	19.4	+10.8
Tiny-ImageNet (300 ep)	7.8	11.1	+3.3	13.7 [†]	14.0	+0.3

[†] For Model A with SIGReg, the default $\lambda_{\text{sig}}=0.1$ underperforms on CIFAR-100 and catastrophically collapses on Tiny-ImageNet (1.8%, 0/3 seeds stable). The Tiny-ImageNet entry uses the rescued $\lambda_{\text{sig}}=0.01$ variant (13.68%, 3/3 seeds stable; Sec. 5.4, Table 4). The CIFAR-100 entry retains the default- λ value because we did not run a low-weight rescue at that scale; this likely understates Model A’s true performance and is a conservative choice. Model B uses the default $\lambda_{\text{sig}}=0.1$ throughout because tangent-space SIGReg operates on a different distribution and does not exhibit the collapse.

Model B specifics. Poincaré curvature $c=1.0$ (except in the curvature ablation of Sec. 5.6). Radius clip $r_{\text{max}}=1.0$. Loss power $p=2$.

Evaluation. We report frozen-encoder linear-probe accuracy on the val set. For Model B we also compute a tangent-space linear probe (features lifted via $\log_0^c$); in every run we measured this we found the Euclidean and tangent probes agree within 0.5 pp, so we report Euclidean probes throughout.

Seeds and statistics. Three seeds per cell (42, 123, 7). Tables report per-seed values and unweighted means. We do not report standard deviations for individual cells because with $n=3$ they are unreliable; instead, where a claim depends on consistency we report whether the claim holds on *every* seed.

Compute. All runs on an academic SLURM cluster with a mix of NVIDIA A5000, A6000, and L40S GPUs, one GPU per run. Empirical walltimes: CIFAR-10 50 ep $\approx$ 10 min, CIFAR-100 200 ep $\approx$ 40 min, Tiny-ImageNet 300 ep $\approx$ 4–5 h, Tiny-ImageNet 600 ep $\approx$ 8–10 h.

5 Results

5.1 Main matrix: Poincaré predictor wins universally

Table 1 gives the primary result: the Poincaré predictor outperforms the Euclidean baseline in every one of the twelve dataset- $\times$ -regularizer cells.

Model B wins every cell, by margins ranging from +0.3 pp (Tiny-ImageNet, SIGReg, with the λ -rescued A baseline) to +10.8 pp (CIFAR-100, SIGReg). The mean improvement across the twelve cells is +4.9 pp. Seed-level consistency varies by cell: in the SIGReg columns of CIFAR-10 and CIFAR-100, every B seed beats every A seed in the same cell, and on Tiny-ImageNet with VICReg every B seed also beats every A seed; the VICReg columns of CIFAR-10 and CIFAR-100 show one or two A seeds reaching into the B range (so the cell-mean comparison still favours B but the per-seed orderings overlap), and the Tiny-ImageNet SIGReg cell (with the λ -rescued A baseline) shows narrow seed overlap consistent with the small +0.3 pp gap. The headline claim is therefore the consistency of the sign across all twelve cell means, not a uniform per-seed dominance; the cell-level magnitude grows with compute (Sec. 5.2).

5.2 Scale-dependent regularizer choice

With the main claim established, we focus on the best column of Table 1 (Tiny-ImageNet, Model B) and ask how the two regularizers scale with compute. Table 2 shows the 300 ep vs. 600 ep comparison.

The gap between B+SIGReg and B+VICReg widens from +2.9 pp at 300 epochs to +4.8 pp at 600 epochs. Seed separation is clean at 600 epochs: the *lowest* B+SIGReg seed (15.71) is above the *highest* B+VICReg seed (12.63). VICReg is essentially flat between 300 and 600 epochs (+0.8

Table 2: **Scaling on Tiny-ImageNet.** Model B linear-probe accuracy (%) at 300 vs. 600 epochs, per seed and mean. SIGReg gains +2.8 pp from the extra compute while VICReg plateaus.

Config	s42	s123	s7	Mean
B+VICReg 300 ep	10.84	11.25	11.35	11.15
B+VICReg 600 ep	10.85	12.63	12.32	11.93
B+SIGReg 300 ep	13.40	14.79	13.75	13.98
B+SIGReg 600 ep	17.60	17.01	15.71	16.77

Table 3: **Negative scaling of Model A.** A+VICReg on Tiny-ImageNet degrades with extra compute, on every seed. Rank and mean-variance remain healthy (~ 161 and ~ 0.4 respectively), so this is not a collapse event but a genuine feature-quality degradation under continued training.

Config	s42	s123	s7	Mean
A+VICReg 300 ep	8.64	6.58	8.28	7.83
A+VICReg 600 ep	4.76	5.86	6.06	5.56
Δ	-3.88	-0.72	-2.22	-2.27

pp), while SIGReg gains +2.8 pp over the same compute, consistent with the hypothesis that VICReg’s first-and-second-moment objective saturates while SIGReg’s distributional matching still has headroom.

5.3 Negative scaling of Euclidean JEP A

For comparison, Table 3 extends the 300 ep-vs-600 ep experiment to the Euclidean baseline.

Every seed regresses; the mean drops by 2.3 pp. Collapse metrics (effective rank ~ 160 out of 192; mean per-dimension variance ~ 0.4) are healthy and similar between the 300 ep and 600 ep runs, so the degradation is not an anti-collapse failure but a genuine feature-quality issue. Combined with Table 2, the compute-trajectory story is unambiguous: Model B gains from extra training on Tiny-ImageNet while Model A loses.

5.4 Ambient SIGReg fails due to regularizer weight, not architecture

The $\dagger$ -marked entries in Table 1 (A+SIGReg at default $\lambda_{\text{sig}}=0.1$) show a near-collapse on Tiny-ImageNet. One plausible conclusion is that ambient SIGReg fundamentally does not work on small ViTs at our scale. We run a two-variant hyperparameter sweep to test whether the issue is the learning rate or the regularizer strength.

Halving the learning rate improves the mean from 1.80 to 4.75 but only one of three seeds becomes stable (the other two still exhibit rank collapse, with effective rank below 90). Reducing the regularizer weight by $10\times$ instead fixes all three seeds at $\approx 13.7\%$, with effective rank ~ 137 and mean-variance ~ 0.23 , stable and consistent. We conclude that ambient SIGReg *works* on ViT-Tiny with Tiny-ImageNet, contradicting the initial appearance of architectural failure, but requires a weight calibrated to the scale. The rescued A+SIGReg (13.68% at 300 epochs) is the value used in the Tiny-ImageNet SIGReg cell of Table 1, where Model B with tangent-space SIGReg achieves 14.0%, a narrow +0.3 pp lead at this compute budget. The 600-epoch comparison of Section 5.2 (B+SIGReg reaching 16.8%) shows that this gap widens with compute, but at 300 epochs alone the B-vs-A SIGReg comparison on Tiny-ImageNet is statistically tight; the strong universal claim of Section 5.1 rests on *sign* consistency across cells, not on a large per-cell margin in this particular cell.

5.5 Projector- $\times$ -regularizer interaction

To check that our SIGReg-vs-VICReg comparison is not an artifact of the BN projector, we re-run Model B on Tiny-ImageNet without it. Table 5 reports the comparison.

The projector moves the two regularizers in opposite directions: VICReg works better without it, SIGReg works better with it, and the result is that the “SIGReg wins on Model B at Tiny-ImageNet”

Table 4: **Ambient SIGReg hyperparameter rescue** (Model A, Tiny-ImageNet, 300 epochs). The apparent catastrophic collapse at default $\lambda_{\text{sig}}=0.1$ is a regularizer-weight issue, not a learning-rate issue: halving the LR only partially helps, while reducing λ_{sig} to 0.01 stabilizes all three seeds. “Diverged” indicates that training produced a non-finite loss before the run completed; the mean column averages the two finite seeds.

Variant	s42	s123	s7	Mean	Stable?
Baseline ($\lambda_{\text{sig}}=0.1$, lr 10^{-3})	3.95	<i>diverged</i>	0.95	1.80	0/3
low-lr ($\lambda_{\text{sig}}=0.1$, lr 5×10^{-4})	8.09	3.09	3.08	4.75	1/3
low-wt ($\lambda_{\text{sig}}=0.01$, lr 10^{-3})	13.91	13.62	13.52	13.68	3/3

Table 5: **Projector $\times$ regularizer interaction** (Model B, Tiny-ImageNet, 300 epochs). The BN projector helps SIGReg and hurts VICReg, which flips the winner at this compute budget.

Config	with projector	without projector	Δ_{proj}
B+VICReg	11.15	14.70	−3.55
B+SIGReg	14.00	11.33	+2.67
Winner	SIGReg +2.9	VICReg +3.4	n/a

claim is conditional on the projector at 300 epochs. At 600 epochs with the projector, SIGReg still wins by +4.8 pp (Table 2), so scale eventually dominates. We report this explicitly rather than cherry-picking: our strongest configuration is B+SIGReg with the BN projector at 600 epochs.

5.6 Curvature ablation

Finally, Table 6 varies the Poincaré curvature c of Model B over an $8 \times$ range.

All four curvatures sit within a 1.0 pp band. The best mean is at $c=0.5$ (15.0%), the main-table choice $c=1.0$ is the lowest (14.0%), and all four are within seed noise. The Poincaré-predictor benefit is therefore not sensitive to the specific curvature choice over a range that varies the ball radius by a factor of two.

5.7 Collapse metrics are not a reliable model selector

We track two common anti-collapse diagnostics: the effective rank of the feature covariance matrix [Roy and Vetterli, 2007] and the mean per-dimension variance. In our Tiny-ImageNet runs, B+SIGReg consistently has *lower* effective rank (~ 88) but *higher* mean-variance (~ 0.60) than B+VICReg (~ 142 , ~ 0.09 respectively), yet achieves higher probe accuracy. Neither metric tracks probe accuracy monotonically: two configurations with nearly identical probe accuracy can have rank differing by 70+, and configurations with identical rank can differ by 5 pp in probe. We thus caution against using anti-collapse metrics as a selection criterion in place of a downstream task probe, and include them in our tables only as diagnostic context.

6 Related work

Joint-embedding SSL. Our experiments build on the I-JEPA architecture [Assran et al., 2023], which in turn extends the general JEPA framework [LeCun, 2022]. Earlier feature-space contrastive SSL methods include SimCLR [Chen et al., 2020], BYOL [Grill et al., 2020], SimSiam [Chen and He, 2021], DINO/DINOv2 [Caron et al., 2021, Oquab et al., 2024], and SwAV [Caron et al., 2020]. Pixel-space approaches such as MAE [He et al., 2022] and SimMIM [Xie et al., 2022] reconstruct raw pixels rather than features. Our work is orthogonal: we do not change the JEPA architecture or training loop, only the geometry of the prediction loss.

Anti-collapse regularizers. The regularizers we study, VICReg [Bardes et al., 2022] and SIGReg [Balestriero and LeCun, 2025], follow a line of work that includes Barlow Twins [Zbontar et al., 2021] and W-MSE [Ermolov et al., 2021]. LeJEPA’s characterization of the isotropic Gaussian

Table 6: **Curvature ablation** (Model B, B+SIGReg+projector, Tiny-ImageNet, 300 epochs). All four curvatures produce means within 1.0 pp of each other; $c=1.0$ is a reasonable default, not cherry-picked.

Curvature c	s42	s123	s7	Mean
0.25	15.46	13.97	14.24	14.56
0.5	14.70	15.68	14.59	15.00
1.0 (main table)	13.40	14.79	13.75	13.98
2.0	14.83	13.96	15.91	14.90

as the optimal anti-collapse distribution provides the theoretical grounding for our tangent-space formulation. Neither VICReg nor SIGReg was originally studied with hyperbolic embeddings; the tangent-space variant we propose is, to our knowledge, new.

Hyperbolic representation learning. Nickel and Kiela [2017] demonstrated that hyperbolic embeddings capture hierarchical structure more efficiently than Euclidean ones. Subsequent work spans two design points: (i) full-network hyperbolic models, where every layer operates on the manifold (hyperbolic neural networks [Ganea et al., 2018, Shimizu et al., 2021], Poincaré ResNet [van Spengler et al., 2023], and fully hyperbolic transformers [Chen et al., 2022]), and (ii) Euclidean encoders with a hyperbolic head or loss, including hyperbolic image classifiers [Khrulkov et al., 2020] and hyperbolic contrastive SSL [Yue et al., 2023, Ge et al., 2023]. Our work falls in the second design point. The closest prior work in spirit is Yue et al. [2023], who use a Euclidean encoder with an exponential map at the output and a hyperbolic contrastive loss; we differ in (i) targeting masked-feature JEPA rather than contrastive SSL, and (ii) composing the hyperbolic loss with a distributional anti-collapse regularizer (tangent-space SIGReg) rather than relying on contrastive negatives. Concurrent work GeoWorld [Zhang et al., 2026] also studies a Poincaré-ball formulation of JEPA, but for goal-conditioned video planning on instructional video benchmarks rather than image pretraining, and without the regularizer comparison, scaling analysis, or projector ablation we report.

Scaling studies of SSL. Goyal et al. [2022], Oquab et al. [2024], and the LeJEPA paper [Balestriero and LeCun, 2025] all report positive scaling of SSL methods under more data or compute. Our observation of *negative* scaling for vanilla I-JEPA on Tiny-ImageNet ViT-Tiny (Sec. 5.3) is unusual and we do not claim it extrapolates to larger backbones; we report it as a reproducible finding at this operating point.

7 Discussion and limitations

Why does a Poincaré predictor help? We do not claim a definitive mechanism. The hyperbolic-embedding literature argues that hierarchical structure (part-whole decompositions, subclass relationships) embeds more efficiently in constant-negative-curvature space. Masked-feature prediction implicitly demands that the predictor recover a target from a related context, which matches the “parent encodes child” structure that Poincaré embeddings naturally express. But our experiments do not test this mechanism directly; we leave the interpretability analysis (e.g., whether target patches cluster closer to the ball origin than context patches, whether the learned distances correlate with WordNet hierarchy) to future work.

Why does SIGReg scale better than VICReg? A concrete hypothesis: VICReg’s per-dimension variance hinge becomes non-binding once variance exceeds γ , at which point only the covariance term contributes gradient. SIGReg, by matching a full distribution rather than moments, has no such saturation point. The +2.8 pp gain of B+SIGReg from 300 to 600 epochs vs. +0.8 pp for B+VICReg is consistent with this story but does not prove it.

Limitations. *Scale.* All experiments use ViT-Tiny (192 dim), and the largest dataset is Tiny-ImageNet. We do not claim the findings extrapolate to ViT-S/ViT-B or to ImageNet-1k. In particular, the “negative scaling of Model A” finding in Table 3 could plausibly reverse at larger scale or on a larger dataset, and the B+SIGReg-vs-VICReg gap at 600 epochs might not extend indefinitely. *Seeds.* Three seeds per cell is the minimum defensible number and we use seed separation rather than

variance estimates to argue consistency. *Curvature.* We use a single shared c across the Poincaré loss and tangent-SIGReg; decoupling these (different curvatures for loss vs. regularizer) might improve results further. *Projector confound.* The projector- $\times$ -regularizer interaction (Sec. 5.5) means our SIG-vs-VIC comparison at 600 epochs is entangled with the projector choice; a no-projector 600-epoch matrix would resolve this but was beyond our compute budget.

Future work. Scaling the study to ViT-S and ImageNet-1k is the obvious next step. Beyond that, two mechanism-level studies are of interest: (i) whether the learned Poincaré geometry recovers known hierarchical structure (WordNet, HDBSCAN-derived clusters), and (ii) whether the tangent-space SIGReg formulation generalizes to other non-Euclidean SSL settings (spherical, product manifolds).

8 Conclusion

We asked a narrow empirical question: does curving only the prediction loss of I-JEPA, while leaving everything else Euclidean, produce better representations? Across a $2 \times 2 \times 3$ matrix of predictor-geometry $\times$ regularizer $\times$ dataset, the answer was unambiguously yes: twelve for twelve. Follow-up experiments showed that at larger compute on Tiny-ImageNet, the hyperbolic predictor additionally composes with a tangent-space version of SIGReg to open a widening gap over VICReg, and that the Euclidean baseline exhibits negative scaling under the same compute increase. We reported the confounds honestly, including a projector- $\times$ -regularizer interaction that affects the 300-epoch ordering, and resolved an apparent failure mode of ambient SIGReg as a hyperparameter issue rather than an architectural one. A one-line change to the loss function is worth measuring before committing to heavier architectural interventions.

Reproducibility. Configuration files, submission scripts, per-epoch logs, and checkpoints for every run reported in this paper are released at [https://github.com/\[anonymized\]/hyperbolic-jepe](https://github.com/[anonymized]/hyperbolic-jepe). Each row of Tables 1–6 corresponds to a single config file and three seeds (42, 123, 7).

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